

Math 141, F11

Quiz 11

Section:

Name: ANSWER

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (4 points) Use Part I of the Fundamental Theorem of the Calculus to find the derivative of the following functions:

$$(a) g(x) = \int_1^x \frac{1}{t^3 + 1} dt$$

$$g'(x) = \frac{1}{x^3 + 1}$$

$$(b) h(x) = \int_1^{x^2} t \ln t dt$$

$$\begin{aligned} h'(x) &= x^2 \ln x^2 \cdot \frac{d}{dx}[x^2] \\ &= (x^2 \ln x^2) \cdot 2x \\ &= 2x^3 \ln x^2 \\ &= 4x^3 \ln x \end{aligned}$$

Problem 2. (6 points) Evaluate the following definite integrals:

$$(a) \int_0^2 (x^3 - 2x) dx$$

$$= \left[\frac{x^4}{4} - x^2 \right] \Big|_{x=0}^{x=2}$$

$$= \left[\frac{16}{4} - 2^2 \right] - [0]$$

$$= 0$$

$$(b) \int_{\pi/2}^{\pi} 2 \sin \theta d\theta$$

$$= [-2 \cos \theta] \Big|_{\theta=\pi/2}^{\pi}$$

$$= (-2 \cos \pi) - (-2 \cos \frac{\pi}{2})$$

$$= 2 - 0 = 2$$